

Augmented Reality based Surgical Navigation for Percutaneous Endoscopic Transforaminal Discectomy

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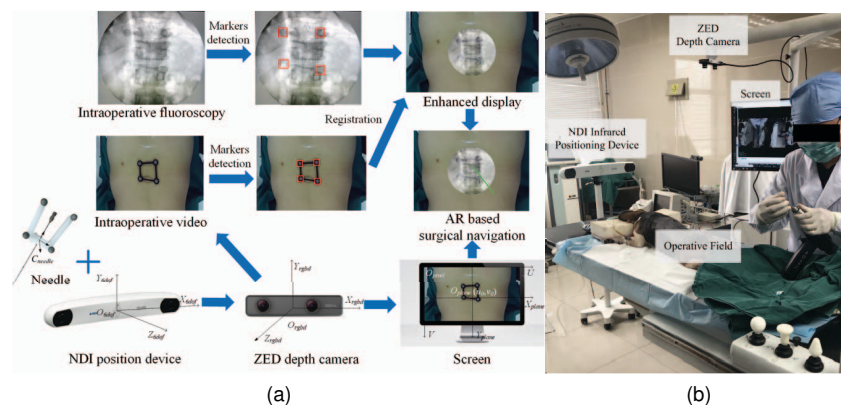


Figure 1: Overview of our developed AR surgical navigation system for percutaneous endoscopic transforaminal discectomy. (a) The main components and structure of system; (b) The environment of operating theatre in vivo animal experiment.

ABSTRACT

Fluoroscopic guidance is a critical step for the puncture procedure in percutaneous endoscopic transforaminal discectomy (PETD). In this paper, we propose an AR surgical navigation system for PETD based on multi-modality imaging information, which contain fluoroscopy, optical tracking and depth camera. We also present a self-adaptive calibration and transformation method between 6-DOF optical tracking device and depth camera, which are in different coordinate systems. With substantially reduced frequency of fluoroscopy imaging, the system can accurately track and superimpose the virtual puncture needle on fluoroscopy images in real-time.

Index Terms: AR based surgical navigation; PETD; Marker detection; Tracking; Registration

In recent years, computer aided orthopaedic surgery has emerged as a new and significant research area, due to the importance of musculoskeletal diseases treatment in orthopaedics [1]. However, most of them are based on the invasive markers, and some of them suffer from low accuracy [2]. Due to the high radiation risk of intraoperative CT, surgeons always choose multiple 2D fluoroscopy imaging to guide the puncture and localization procedures in PETD. So the 3D orthopedic surgery navigation techniques based on CT cannot be used directly. In this paper, we propose an AR-based sur-

gical navigation system for PETD. Using an optical tracking device to capture 3D motion (6-DOF) of the surgical tool (puncture needle), and a depth camera to obtain intraoperative video, this system can accurately track the puncture needle in real-time and superimpose the virtual needle on fluoroscopy images fused with intraoperative video (Fig.1).

1 SYSTEM DESCRIPTION

Our AR based surgical navigation system contains five major hardware components: a NDI infrared positioning device (Polaris Spectra), a puncture needle attached with four markers, a custom-made non-invasive fiducial with four markers, two ZED depth cameras and a workstation with screen. In our system, NDI infrared positioning device can track the 6-DOFs motion of puncture needle with the root mean square error (RSME) limited to 0.25mm. Two ZED depth cameras are employed to capture the intraoperative videos, which will be registered and fused with the frontal and lateral fluoroscopic images respectively during navigation. The fiducial markers, which is usually attached on the patient back skin during puncture procedure, is utilized for the registration between fluoroscopic image and intraoperative video.

The workflow of our system consists of two stages. The first is the initialization. Based on recognition algorithm for non-invasive markers, besides the fluoroscopic image can be registered with the intraoperative video, we can acquire the 3D position of fiducial markers in both infrared positioning device space (6-DOF) and depth camera space (RGB-D). Then the coordinates transformations between different imaging modalities can be solved by singular value decomposition (SVD). Due to the workspace limit of

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NDI infrared positioning device, we need to calibrate the system before navigation to ensure it can track puncture needle robustly. Twenty different positions of fiducial markers are placed and collect their 3D coordinates under both optical tracking device and depth camera. The collected 3D coordinates are used to build the effective navigation space. The second stage is AR navigation and enhanced display. We superimpose the fluoroscopic image on the intraoperative videos by markers detection and registration. Then the puncture needle, which is tracked from 6-DOF coordinate system, is projected as "virtual needle" in screen space to achieve the intraoperative navigation.

1.1 Registration between fluoroscopic image and the intraoperative video

Here we design a lightweight non-invasive fiducial with markers and present a straightforward and effective method to detect these markers. In registration, the marker with green background is named as Marker₀, and the marker with red background is named as Marker₁. All these sphere markers can be captured by infrared positioning device to get the 3D position and orientation of fiducial markers (6 degrees of freedom) in real-time. The fiducial is pasted on the back skin of patient near the area of lumbar vertebra. Considering there is seldom obvious displacement of patient's lumbar vertebra during puncture and localization procedures in PETD, this non-invasive fiducial can also be used for the registration between fluoroscopic image and the intraoperative video. We train a neural network to detect the non-invasive markers in both intraoperative video and fluoroscopic images based on YOLO-v3 [3]. In the end, the trained model is utilized to detect and acquire the position of markers in the video. For registration, four markers are tracked and located to obtain the position and scale of their boundingboxes in the intraoperative video and fluoroscopic images during surgery. Considering the spatial relationship between intraoperative video and fluoroscopic imaging, we treat the spatial transformation from fluoroscopic image space to video space as the similarity transformation.

1.2 Self-adaptive calibration and transformation between different modalities

We present a self-adaptive calibration and transformation method between different imaging modalities. It can build a reasonable transformation among different dimension and DOF coordinate systems, and accurately project the 6-DOF motion of puncture needle captured by optical tracking device into the 2D screen space. It overcomes the limitation of the existing surgical navigation systems on the location of navigation equipment, and provides remarkable convenience for surgeons in operation. Then we normalize the 3D coordinates of N markers pair to make the coordinate systems of infrared positioning device and the depth camera have the same scale. After calibration, according to the rotation matrix $\mathbf{R}$ and the translation matrix $\mathbf{T}$, we can map the real-time updated coordinates $\mathbf{M}(x, y, z, r_x, r_y, r_z)$ of the object (markers attached to the puncture needle) in the infrared positioning device space to the coordinates $\mathbf{M}(\hat{x}, \hat{y}, \hat{z})$ in the depth camera space during navigation. Then the coordinates $\mathbf{m}(u, v)$ of the object in the image plane can be obtained by the perspective projection transformation. In the navigation system, we use a colour line segment as the "virtual needle" to enhanced display the localization of real puncture needle in the frontal and lateral fluoroscopic images, which can be superimposed on the intraoperative videos.

2 EXPERIMENTS AND VALIDATION

Fig.2 demonstrate three key frames from the AR navigation videos with the "virtual needle" superimposed on the frontal fluoroscopic image in animal experiments. The positioning error of the needle tip is about 1.8mm (lower than 2mm), which meets the clinical requirements. The second experiment is the accuracy measurement of our AR surgical navigation system. Tab.1 and 2 illustrate the experimental results, in which the average positioning error is 1.98mm

with a standard deviation of 0.41mm and the average orientation error is 1.19° with a standard deviation of 0.69°. Moreover, our system significantly reduces the frequency of fluoroscopy imaging and lower the radiation risk for both the patient and surgeons.

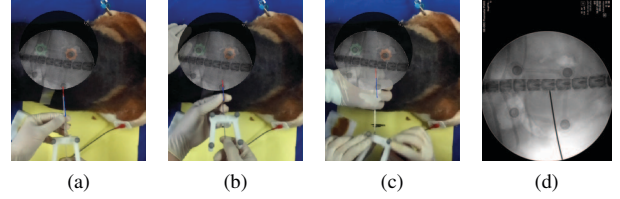


Figure 2: The puncture procedure with AR navigation. (a) The beginning of puncture (frame 157); (b) Puncture in progress (frame 1874); (c) The end of puncture (frame 4829); (d) The frontal fluoroscopic image with the needle after puncture procedure.

Table 1: Positioning error (mm) of the needle tip for AR surgical navigation on in vivo experiments.

Data	Mean±SD	Min	Max
Test 1	2.29±0.41	1.67	2.78
Test 2	2.05±0.48	1.56	2.76
Test 3	1.77±0.27	1.53	2.21
Test 4	1.79±0.49	1.11	2.36
All	1.98±0.41	1.11	2.78

Table 2: Orientation error (degree) of the puncture needle for AR surgical navigation on in vivo experiments.

Data	Mean±SD	Min	Max
Test 1	1.10±0.73	0.30	2.10
Test 2	1.43±1.12	0.20	2.90
Test 3	1.20±0.56	0.40	1.90
Test 4	1.05±0.35	0.50	1.40
All	1.19±0.69	0.20	2.90

3 CONCLUSION AND FUTURE WORK

This paper presents an AR surgical navigation system for PETD based on fluoroscopy guidance. We design a lightweight non-invasive fiducial with markers to assist the registration between the fluoroscopic images and intraoperative videos. We believe that this system still has room for improvement. In future, we will increase the system computation speed with CUDA to make the navigation more smooth. We also plan to improve the calibration process which is not convenient at present. The distribution of the fiducial markers influence the effective spatial scope of navigation system workspace.

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